

Environmental variability and bet-hedging

Learning objectives

1. Connect environmental variability to changes in lambda
2. Compare populations with a range of growth rates and sensitivities to environmental variability
3. Model a population with a seed bank

last lecture: unrestricted population growth

modeled demographic stochasticity by making lambda continuous random variable

this is a workaround for modeling dynamics where temporal variable affects rates

Temporal environmental variability

- example: climate
- $\lambda > 1$ in wet years
- $\lambda < 1$ in dry years

Spatial environmental variability

on small scales, patchiness makes populations grow or shrink

- example: light & understory species
- λ varies in space
- lattice models, cells affect population growth

draw 4x4 lattice, random patches and plants

lambda=0.9 or 1.1 to indicate where populations growing and shrinking

Bet-hedging

- variability is a selective agent
- populations vary sensitivity to environmental variability

go back to modeling lambda as continuous random variable

Population 1:

$$\lambda_1 \sim \text{Normal}(\mu = 1.2, \sigma = 0.5)$$

Population 2:

$$\lambda_2 \sim \text{Normal}(\mu = 1, \sigma = 0.2)$$

- pop 1 is more sensitive than pop 2 *indicated by sigma*
- $N_1 < N_2$ even though $\lambda_1 > \lambda_2$

how do plants actually do this? with seed banks

Seed banks

- seed germination = sensitivity to good vs. bad years (σ)

with a seed bank, favorable years seeds germinate and become adult plants, stressful years seeds germinate and die or don't germinate

- low sensitivity: similar number of seeds germinate each year
- high sensitivity: number depends on environmental cues, boom-bust
- spread risk *naturally seeds will germinate in different years, even if they experience same exact environment year to year*

Modeling seed banks

in lab, we'll model density independent growth with a seed bank

still use λ and N for per capita growth rate and population size

here are some of the terms and a quick intro to the model

- g = germination rate of seeds produced by adults
- s = survival of seeds in seed bank

$$N_{t+1} = N(1 - g)s + \lambda gN$$

- includes the seeds that go into the seed bank
- first term = seeds in seed bank
- second term = new seeds